Searching for Compact Algorithms: CGEN

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Abstract. In this paper we describe an AES-like pseudo-random number generator called CGEN. Initial estimates suggest that the computational resources required for its implementation are sufficiently modest for it be suitable for use in RFID tags.

1 Introduction

There has been much recent interest in extending cryptographic techniques to resource-constrained devices. Much of this work has centered around the design of exotic primitives and protocols suitable for demanding environments such as *radio frequency identification* (RFID) tags. There is, however, something unsettling about building a security infrastructure on entirely new design techniques that might be vulnerable to innovative attack. However, at the same time, many standardised algorithms are not suited to deployment on the cheapest tags. This gap between extreme proposals and more conventional primitives has made the design of lightweight, but trusted, algorithms an active topic of research.

In this paper we propose an algorithm that is designed to (partially) fill this gap and we introduce a *pseudo-random number generator* (PRNG) that we call CGEN for "compact generator". Some reasons for concentrating on this type of primitive are provided in Section 1.1, but our goal is a PRNG that would occupy 1000-1500 gate equivalents (GE) in silicon, make modest power demands, and yet fit this middle ground of *dedicated-but-strong* cryptography.

The properties we seek for a PRNG essentially coincide with those we expect for a good stream or a block cipher in an appropriate mode. Unfortunately, however, there are currently no widely-trusted stream ciphers that are suitable for compact hardware implementation, though there are efforts to try and rectify this with the eSTREAM initiative [11]. Two proposals in particular—the more traditional GRAIN [16] and the innovative TRIVIUM [5]—appear to offer considerable promise. By contrast, we do have a well-trusted block cipher—the AES [21]—and, up to a number of encryptions dictated by birthday attacks, the output from this block cipher in counter mode would be adequate for our purposes. However, even when considering recent impressive implementation results [12], the hardware requirements of the AES remain excessive.

In this paper we build on recent trends in block cipher design but, for the application we have in mind, we observe that a block cipher offers more than we

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need. In fact the restricted scope for attacks against the proposed PRNG, particularly when used in its intended environment, suggest that much of the usual block cipher machinery can be removed. Our goal, therefore, is an algorithm whose dedicated purpose[1] is to act as a PRNG and our proposal is best viewed as a complement to algorithms like GRAIN and TRIVIUM, but one that uses typical block cipher techniques in its design.

1.1 Why a (New) Compact Pseudo-random Number Generator?

The recent surge of interest in computationally-restricted devices has generated many proposals for a wide range of protocols. Often, a hidden requirement is that there be an efficient way to generate random bits, but it is not always clear how this might be best achieved and a tag-specific PRNG may be more practical. Another reason for looking at a PRNG can be found when considering public-key schemes such as the GPS identification scheme [14,22]. A common optimisation in many commitment-challenge-response schemes is the use of *coupons*. These consist of pre-computed commitments that are stored securely on the tag and used one-by-one. Depending on storage limitations and the specifics of the scheme, a further optimisation can sometimes be attained by regenerating (parts of) the coupon with an efficient and compact PRNG as specified in ISO-9798 [15]. Thus, a compact PRNG might be used to generate a modest number of coupons for use within a larger scheme. More generally, for a variety of constrained applications, the ability to efficiently generate pseudorandom bits from a fixed, per-device, secret seed could well be of great interest.

Block ciphers, or constructions built around block ciphers [17], can provide a convenient solution and a benchmark is provided by the AES for which a compact implementation requires around 3600 GE [12]. Despite the name, the *Tiny Encryption Algorithm* TEA, and extensions [25,26], require more resources than we would like, and our attention shifts to two dedicated block cipher proposals mCRYPTON [18], which has many design features in common with CGEN and will be discussed in Sections 2.2 and 3.2, and SEA, the *Scalable Encryption Algorithm* [24] which achieves a small implementation footprint at some cost to performance. Other approaches to designing a PRNG might rely on the use of dedicated hash functions such as MD5 [23] or SHA-1 [20]. But independently of issues about the suitability of these primitives for new applications, the resource cost of implementing such algorithms is high. We therefore believe that a PRNG that stays close to established design principles, but provides the necessary functionality at a reduced cost, may be attractive.

In general terms there might be two approaches to the design of a new PRNG. We might try and develop a "provably secure" scheme that would relate any advantage in predicting the output from the PRNG to an advantage in solving some underlying hard problem. While such an approach has typically been viewed as difficult in symmetric cryptography, particularly when reasonable performance

[1] Since CGEN is essentially built around a block cipher one could use it in this way after some adjustments.